Calculus & Mechanics Comprehensive Review

SUBJECT: MATHEMATICS • DATE: SUN, NOV 1, 2026 • VEYLENTRA ACADEMIC OS

AP Calculus Lecture Notes

Core Differential Rules

- Derivatives define instantaneous rates of change.
- Chain rule applies to nested compositions: $f(g(x))' = f'(g(x)) \cdot g'(x)$.

Comparative Reference Table

Rule Name	Formulation	Operational Domain	Status
Power Rule	d/dx(x^n) = n*x^(n-1)	All Real Numbers	Verified
Product Rule	(uv)' = u'v + uv'	Continuous Functions	Verified
Quotient Rule	(u/v)' = (u'v - uv')/v^2	Non-zero Denominator	Verified
L'Hôpital	lim f/g = lim f'/g'	Indeterminate 0/0 or inf/inf	Verified

Task List

- [x] Complete Problem Set 5
- [ ] Review trigonometric substitution proofs
- [ ] Attend Tuesday seminar on Fourier series

> Ø=Ü¡ K e y T a k e a w a y : A l w a y s c h e c k t h a t t h e i n d e t e r m i n a t e c o n d i t i o n h o l d s b e f o r e

```typescript
function derivative(f: (x: number) => number, x: number, h = 1e-7): number {
  return (f(x + h) - f(x)) / h;
}
```

End of lecture notes review.